Causality and Dealing with Omitted Variables Bias

These notes accompany Lecture 7. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Lecture 6 dealt with heteroskedasticity and non-normality. Those are real problems, but relatively minor ones: they do not bias or make OLS inconsistent. This lecture takes on the serious case – correlation between the error u and one or more explanatory variables, which does both.

Learning outcomes – after this lecture you should be able to:

1. Explain why functional form misspecification is a special case of omitted variable bias, and test for it with RESET.
2. Use a proxy variable to reduce omitted variable bias, and state the two conditions the plug-in solution requires.
3. Define instrument exogeneity and instrument relevance, and explain why a good proxy is a bad instrument.
4. Describe what each stage of 2SLS does, and interpret its output.
5. Test for endogeneity, for overidentifying restrictions, and for weak instruments.
6. State the main criticisms of IV estimation.
7. Use pooled cross sections and the difference-in-differences estimator to evaluate a policy change.

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1. Functional form misspecification

A multiple regression model suffers from functional form misspecification when it does not properly account for the relationship between the dependent variable and the observed explanatory variables.

> **Which assumption is this about?**
>
> Not MLR.1. That assumption requires linearity in the parameters, not in the variables – and it is satisfied by quadratics, logs and interactions alike.
>
> This is about MLR.4, the zero conditional mean. Functional form misspecification is a special case of omitted variable bias, in which the omitted variable is a function of an explanatory variable already in the model.
>
> That is why it biases the estimates: leaving out $educ^2$ is, formally, leaving out a relevant regressor that is correlated with educ.

If hourly wage is genuinely determined in logs and we estimate it in levels, or if an interaction term belongs in the model and we omit it, we are committing functional form misspecification – and this generally leads to biased and inconsistent estimators of the β_j.

1.1 Detecting it informally

It often makes sense to add quadratic terms of any significant variables and perform a joint test of significance. If the quadratics are significant they can be added to the model, at the cost of complicating its interpretation.

But be careful reading the result. Significant quadratic terms can be symptomatic of a different functional form problem – using the level of a variable where the logarithm is appropriate, or the reverse. It can be difficult to pinpoint the precise reason a functional form is wrong. Fortunately, in many cases logarithms and quadratics are enough to capture the important non-linear relationships in economics.

1.2 The RESET test

Ramsey's (1969) regression specification error test formalises the idea. The logic is simple: if the original model satisfies MLR.4, then no non-linear function of the independent variables should be significant when added to the equation.

1. Estimate the original model and obtain the fitted values $\hat{y}$.
2. Re-estimate the model with powers of the fitted values added – the squared and cubed terms have proven useful in most applications.
3. Test the joint significance of the added terms. H_0: their coefficients are all zero, i.e. no functional-form misspecification and no omitted non-linear terms.
4. Reject H_0 and there is evidence of misspecification: omitted non-linear terms, incorrect functional form, or omitted variables correlated with the regressors.

In EViews: View → Stability Diagnostics → Ramsey RESET Test.

What RESET can and cannot do

RESET does not tell you what is wrong – only that something is.

And passing RESET is not a clean bill of health. The test checks for missing non-linear combinations of the regressors you already have. It says nothing about a genuinely omitted variable that is not a function of them.

If u is correlated with an explanatory variable for any reason, that variable is endogenous and the estimates are biased – whatever RESET reports.

2. Proxy variables for unobserved regressors

2.1 The problem

Consider a wage equation that explicitly recognises that ability affects wages:

$$log(wage) = \beta_0 + \beta_1\, educ + \beta_2\, exper + \beta_3\, ability + u$$

We want to hold ability fixed when measuring the return to education. Ability is unobserved, so it goes into the error term – and because education is correlated with ability, the OLS estimator of β_1 (and β_2) is biased.

One remedy is a **proxy variable**: something observed that is related to the unobserved variable we would like to control for. In the wage equation, IQ is a natural candidate for ability. Note what this does *not* require – IQ need not be the same thing as ability, only correlated with it.

2.2 The plug-in solution

Write the model with three regressors, two observed:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3^* + u$$

where x_3^* is unobserved and x_3 is an observed proxy for it. The proposal is simply to pretend the two are the same and regress y on x_1, x_2 and x_3 – hence the plug-in solution.

Consistency of $\hat{\beta}_1$ and $\hat{\beta}_2$ then requires two assumptions:

- **u is uncorrelated with x_3.** The proxy is irrelevant in the population model once x_1, x_2 and x_3^* are included – it earns its place only through its relationship with x_3^*.
- **Once x_3 is controlled for, $E(x_3^*)$ does not depend on x_1 or x_2.** In the wage equation: the average level of ability varies with IQ, but not further with education and experience.

IQ as a proxy for ability

	Return to education	Reading
Model 1: no proxy	6.5%	Too high, if omitted ability is positively correlated with educ
Model 2: IQ included	5.4%	Falls, exactly as omitted-variable-bias reasoning predicts

IQ itself has a statistically significant positive effect: everything else equal, ten more IQ points is predicted to raise monthly earnings by about 3.6%. Education still plays an important role, though a smaller one than originally estimated.

> **A common objection, and the answer**
>
> Adding IQ to a regression that already contains education worsens multicollinearity, inflating the standard errors.
>
> True – but the added multicollinearity is a necessary evil. We are trading precision for a reduction in bias, and bias is the more serious problem, since no sample size cures it.

2.3 Lagged dependent variables as proxies

Sometimes we have no idea how to construct a proxy for the omitted factor. One general-purpose device is to include the dependent variable from an earlier period:

$$crime = \beta_0 + \beta_1\, unem + \beta_2\, expend + \beta_3\, crime_{-1} + u$$

Why include the lagged crime rate? Because cities with high historical crime rates spend more on crime prevention. The unobserved factors that drive crime are therefore correlated with expend – so a pure cross-sectional regression will not recover the causal effect of law enforcement spending.

Including $crime_{-1}$ permits a sharper thought experiment: if two cities have the same previous crime rate and the same current unemployment rate, then β_2 measures the effect of another dollar of law enforcement.

Estimated on 46 cities for 1987, using the 1982 crime rate: without the lagged rate, the effects of unemployment and expenditure are counterintuitive and neither is significant. Adding log crime from five years earlier changes the coefficients substantially. A 1% higher crime rate in 1982 predicts about a 1.19% higher rate in 1987.

Putting in a lagged y as a general control for unobservables is hardly perfect – but it can help produce a better estimate of the effect of a policy variable.

Dependent variable: log($crmrte_{87}$)

Independent Variables	(1)	(2)
$unem_{87}$	−.029 (.032)	.009 (.020)
log($lawexpc_{87}$)	.203 (.173)	−.140 (.109)
log($crmrte_{82}$)	—	1.194 (.132)
intercept	3.34 (1.25)	.076 (.821)
Observations	46	46
R-squared	.057	.680

Source: Wooldridge (2013).

3. Instrumental variables and two stage least squares

The proxy approach can work well, but a good proxy is not always available. Instrumental variables take a different route: leave the unobserved variable in the error term, and use an estimation method that recognises it is there.

3.1 What an instrument is

Take the simple model $y = \beta_0 + \beta_1 x + u$, where we suspect $\text{Cov}(x, u) \neq 0$. Suppose we can observe a variable z satisfying two conditions:

Condition	Formally	In words
Instrument exogeneity	$\text{Cov}(z, u) = 0$	z has no partial effect on y once x and the omitted variables are controlled for, and is uncorrelated with the omitted variables
Instrument relevance	$\text{Cov}(z, x) \neq 0$	z is related – positively or negatively – to the endogenous explanatory variable

Then z is an **instrumental variable** for x.

Why a good proxy is a bad instrument

> **The sharpest distinction in this lecture**
>
> A proxy for ability must be as highly correlated as possible with ability. That is the whole point of it.
>
> An instrument must be uncorrelated with ability. That is the whole point of it.
>
> So IQ is an excellent proxy for ability and a useless instrument for education. The two roles pull in opposite directions, and confusing them is the most common error in this material.

Labour economists have instead used family background variables as instruments for education. Mother's education is positively correlated with a child's education – though it may also be correlated with the child's ability, through inherited ability and early nurturing, which is precisely the worry. Number of siblings while growing up is another candidate: more siblings is typically associated with lower average education.

3.2 IV in the multiple regression model

Consider a model with two explanatory variables, using notation that distinguishes their status:

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1$$

- z_1 (exper) – exogenous, so it may appear as a regressor and serve as its own instrument.
- y_2 (educ) – the endogenous explanatory variable, suspected of correlation with u_1.

Estimated by OLS, all the estimators are biased and inconsistent – not only the one on the endogenous variable. Since z_1 already appears in the equation it cannot serve as the instrument for y_2, so we need a further exogenous variable z_2 that does not appear in the equation.

Because z_1 is present, relevance must now be stated as a partial correlation. Write the endogenous regressor as a function of all the exogenous variables:

$$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v_2$$

The key identification condition is $\pi_2 \neq 0$: after partialling out z_1, the instrument and the endogenous regressor must still be correlated.

3.3 Two stage least squares

The intuition is a decomposition. OLS uses all the variation in education. If education is endogenous, part of that variation is correlated with the error – ability, motivation, and so on:

educ = exogenous variation + endogenous variation

To obtain consistent estimators we must use only the first part.

First stage

Regress the endogenous variable on the instrument (and any other exogenous regressors) and keep the fitted values:

$$\widehat{educ} = \pi_0 + \pi_1 fatheduc$$

The decomposition is then exact:

$$educ = \widehat{educ} + (educ - \widehat{educ})$$

Component	Property	Use
êduc (fitted)	Correlated with educ (relevance); uncorrelated with u (exogeneity)	Valid for estimating the causal effect of education on wages
educ – êduc (residual)	Reflects ability, motivation and other unobservables that affect wages directly	Invalid for causal inference – this is the part we discard

Second stage

Regress the dependent variable on the fitted values:

$$log(wage) = \beta_0 + \beta_1 \hat{educ}$$

β_1 can now be interpreted as the causal effect of education on wages.

> **Do not run the two stages by hand**
>
> If you perform the second stage manually, the standard errors and test statistics will be wrong – they take no account of the fact that êduc is itself estimated.
>
> Any software supporting 2SLS asks for the dependent variable, the full list of explanatory variables (endogenous and exogenous), and the full instrument list (all exogenous variables, including those already in the equation). The output looks much like OLS.

3.4 Return to education for married women

Using data on married working women, first in the simple model with father's education as the instrument:

Specification	Return to education	Note
OLS, simple model	10.8%	–
IV with fatheduc, simple model	5.9%	About half the OLS figure – consistent with omitted ability bias
OLS, with exper and exper²	10.8%	–
2SLS with fatheduc and motheduc	6.1%	Barely significant at 5%, because of a relatively large standard error

Relevance is easily checked: a simple regression of educ on fatheduc shows a statistically significant positive relationship. With two instruments, we test their joint significance in the first stage and reject the null that they are jointly insignificant – education is partially correlated with parents' education.

Note the pattern across the table: the IV estimate is consistently about half the OLS estimate, and consistently less precise. Both are exactly what the theory predicts.

3.5 Diagnostics

Overidentification: are the instruments exogenous?

If there are more instruments than endogenous regressors, the model is overidentified and instrument validity becomes partially testable. EViews computes the J-statistic for the null that all instruments are exogenous. In this application the p-value is above any conventional level, so we cannot reject exogeneity.

Note "partially": the test compares the instruments against each other. If all of them are invalid in the same way, the test will not detect it.

Endogeneity: was 2SLS necessary?

2SLS is less efficient than OLS when the regressors are in fact exogenous, so it is worth testing whether it is needed at all. Hausman (1978) suggested comparing the OLS and 2SLS estimates directly and asking whether the difference is statistically significant.

Here the result is not clear-cut: education appears exogenous at the 5% level and endogenous at 10%. That is a signal to work further on the specification rather than to declare the matter settled.

Weak instruments

IV can produce strange results when relevance is nearly violated – low, but not zero, correlation between z and x. In a given application it is hard to say how low is too low; the theoretical literature models the correlation as shrinking to zero at the rate $1/\sqrt{n}$.

The standard diagnostic compares the Cragg-Donald F-statistic against the Stock-Yogo critical values; the null of weak instruments is rejected when the F-statistic exceeds them. Here the Cragg-Donald statistic is 55.4, comfortably above every Stock-Yogo critical value, so instrument relevance is not a concern in this specification.

3.6 Criticisms of IV estimation

1. Questionable instrument validity. Exogeneity is fundamentally untestable in the just-identified case. What if father's education is related to the daughter's ability? We cannot check this directly.
2. Weak instruments. Even a valid instrument may be only weakly correlated with the endogenous regressor. Strong, credible instruments are rare.
3. Loss of efficiency. IV uses only the instrument-induced variation, so standard errors are larger. Results can be statistically insignificant even when the effect is economically meaningful.
4. Sensitivity. Estimates can be highly sensitive to specification and to the choice of instrument.
5. Local rather than average effects. The estimate identifies the effect for the subpopulation whose behaviour actually responds to the instrument, which limits external validity.

When instruments are poor – correlated with the error, weakly correlated with the endogenous regressor, or both – 2SLS can be worse than OLS. Used properly, it lets us estimate causal effects in the presence of endogenous regressors; used carelessly, it produces imprecise and fragile estimates.

4. Policy analysis with pooled cross sections

4.1 Independently pooled cross sections

An independently pooled cross section is obtained by sampling randomly from a large population at different points in time. In each year we might draw a random sample of hourly wages, education and experience from the population of working people in Lithuania.

Pooling raises the sample size, giving more precise estimators and more powerful tests. But note the key feature: the units are different in each year. And unlike a single random sample, observations from different periods are not identically distributed – the distributions of wages and education change over time in most countries.

We reflect this by allowing the intercept to differ across periods, through year dummies for all but one year (usually the earliest as base).

4.2 Changes in the return to education and the gender gap

Pooling 1978 (base year) and 1985, with 550 people sampled in 1978 and a different set of 534 in 1985:

$$log(wage) = \beta_0 + \delta_0\, y85 + \beta_1\, educ + \delta_1\, y85{\cdot}educ + \beta_2\, exper + \beta_3\, exper^2$$

$$+ \beta_4\, union + \beta_5\, female + \delta_5\, y85{\cdot}female + u$$

The interaction terms are what carry the story: δ_1 measures how the return to a year of education changed over the seven years, and δ_5 how the gender gap changed.

Quantity	Estimate	Inference
Return to education, 1978	7.5%	–
Change by 1985 (δ_1)	+1.85 p.p.	Significant at 5%, two-sided

Quantity	Estimate	Inference
Gender gap, 1978	31.7% (27.2% exact)	–
Change by 1985 (δ_5)	gap fell ≈ 8.5 p.p.	Significant at 5% against the positive one-sided alternative

A question worth pausing on: nominal or real wages?

Hourly wage here is in nominal dollars, and nominal wages grow through inflation alone. Should we deflate?

It turns out not to be necessary – provided a 1985 year dummy is included and log(wage) rather than wage is the dependent variable.

Deflating means subtracting log(P85) from the 1985 observations, and that constant is simply absorbed into the coefficient on y85. Every other coefficient is untouched.

4.3 Difference-in-differences

Two cross sections collected before and after an event can be used to determine its effect. Meyer, Viscusi and Durbin (1995) studied the length of time in weeks that an injured worker receives workers' compensation. On 15 July 1980 Kentucky raised the cap on weekly earnings covered by the scheme.

The hypothesis: raising the cap has no effect on the benefit available to low-income workers, but it makes staying on compensation less costly for a high-income worker.

Why the naive comparison fails

A naive analyst would use only the post-policy data and regress duration on a high-earner dummy. The problem is that high earners might differ from low earners in duration for many reasons that have nothing to do with the policy – different injuries, different occupations, different outside options. That comparison confounds the policy effect with a pre-existing group difference.

The estimator

Pooling both periods gives the standard specification:

$$log(durat) = \beta_0 + \delta_0\, afchnge + \beta_1\, highearn + \delta_1\, afchnge{\cdot}highearn + u$$

where *durat* denotes the length of time (in weeks) that an injured worker receives workers' compensation; *afchnge* is a binary variable equal to one for the period after the change, and zero – before the change*; highearn* is a binary variable equal to one if the worker is high-income, and zero otherwise.

Parameter	What it measures
β_0	Average duration for low-income workers before the change
δ_0	Change in duration for low-income workers after the change – the common time trend
β_1	Pre-existing difference between the two groups, not caused by the policy
δ_1	The parameter of interest: the change in the gap between the groups due to the policy

Equivalently, in terms of group means:

$$\delta_1 = (\bar{y}_{2,T} - \bar{y}_{2,C}) - (\bar{y}_{1,T} - \bar{y}_{1,C})$$

where T and C denote treatment and control groups and the subscripts 1 and 2 the two periods. The regression is needed not to obtain the estimate – the group means already give it – but to obtain its standard error.

Results

$$\log(durat) = 1.126 + 0.008 afchnge + 0.256 highearn + 0.191 afchnge * highearn + u$$

	Simple model	With controls
afchnge·highearn (δ_1)	0.191 (p = 0.005)	0.178 (p = 0.006)
afchnge (δ_0)	0.008 (p = 0.86)	0.045 (p = 0.28)

Average time on workers' compensation for high earners rose by about 19% as a result of the increased cap. The coefficient on afchnge is small and insignificant – exactly as expected, since the policy should not have affected low-income workers. Adding controls for the kinds of people and types of injuries changes the interaction estimate very little, which is reassuring.

This is also a good example of obtaining a fairly precise estimate of a policy effect even though the model explains little of the total variation in the dependent variable.

4.4 Natural experiments

A **natural experiment** (or quasi-experiment) occurs when some exogenous event – often a policy change – alters the environment in which individuals, families, firms or cities operate. It always has a **control group**, unaffected by the change, and a **treatment group**, thought to be affected.

To control for systematic differences between the two groups we need two periods, one before and one after. The sample therefore breaks into four cells: control before, control after, treatment before, treatment after. With dT for the treatment group and d2 for the second period:

$$y = \beta_0 + \delta_0\, d2 + \beta_1\, dT + \delta_1\, d2{\cdot}dT + \text{other factors}$$

δ_1 measures the effect of the policy. Without other factors in the regression it is exactly the difference-in-differences estimator; with controls added the OLS estimate no longer has that simple closed form, but its interpretation is the same.

5. Summary

1. Functional form misspecification is a special case of omitted variable bias, in which the omitted variable is a function of an included regressor. RESET detects it but does not diagnose it, and passing RESET does not rule out endogeneity.
2. A proxy variable stands in for an unobserved regressor. Under reasonable assumptions, including it in an OLS regression eliminates or at least reduces the omitted variable bias – at the cost of some added multicollinearity.

3. Instrumental variables leave the unobserved factor in the error and use outside variation instead. An instrument must be exogenous (uncorrelated with u) and relevant (correlated with the endogenous regressor). A good proxy is a bad instrument, and vice versa.
4. 2SLS isolates the exogenous part of the endogenous regressor in a first stage and uses only that variation in the second. Let the software do both stages, or the standard errors will be wrong.
5. Three diagnostics matter: overidentification tests for instrument exogeneity, a Hausman-type test for whether IV was needed, and the Cragg-Donald statistic against Stock-Yogo values for instrument strength.
6. Independently pooled cross sections combine random samples from different periods. Allow different time intercepts, and interact time dummies with key variables to see how relationships have changed.
7. Difference-in-differences compares the change in the treatment group with the change in the control group, netting out both the pre-existing group difference and the common time trend. It is the workhorse of natural-experiment policy evaluation.

> **The thread running through the lecture**
>
> Every method here exists because of one failure – MLR.4, the correlation between a regressor and the error.
>
> The proxy approach tries to observe the missing variable. IV gives up on observing it and finds variation that is clean by construction. Difference-in-differences finds a comparison group that absorbs it.
>
> None of them is a formula you can apply mechanically. Each requires an argument about where the variation came from – which is the same argument Lecture 1 said separates econometrics from data analysis.

Key terms

Term	Meaning
Functional form misspecification	The model fails to capture the true relationship; a special case of omitted variable bias where the omitted term is a function of included regressors.
RESET	Ramsey's test: add powers of the fitted values and test their joint significance.
Endogenous explanatory variable	A regressor correlated with the error term. Causes bias and inconsistency.
Proxy variable	An observed variable correlated with an unobserved regressor, included in its place.
Plug-in solution	Substituting the proxy for the unobserved variable and estimating by OLS.
Instrumental variable	A variable that is uncorrelated with u (exogeneity) but correlated with the endogenous regressor (relevance).
Instrument exogeneity	Cov(z, u) = 0. Untestable in the just-identified case.
Instrument relevance	Cov(z, x) ≠ 0; with other regressors present, $\pi_2 \neq 0$ in the first stage.
Two stage least squares (2SLS)	First stage isolates the exogenous variation in the endogenous regressor; second stage uses only that variation.

Term	Meaning
Overidentification	More instruments than endogenous regressors, allowing a partial test of instrument exogeneity via the J-statistic.
Weak instruments	Low but non-zero correlation between instrument and endogenous regressor. Diagnosed with the Cragg-Donald statistic against Stock-Yogo critical values.
Pooled cross section	Independent random samples drawn from the same population at different times; different units each period.
Natural experiment	An exogenous event creating a treatment group and an unaffected control group.
Difference-in-differences	The change in the treatment group minus the change in the control group; the coefficient on d2·dT.

Review questions

Use these to check your understanding before the seminar.

1. Explain why functional form misspecification is a violation of MLR.4 rather than MLR.1.
2. Your model passes RESET at the 10% level. What have you learned, and what have you not learned?
3. State the two conditions the plug-in solution requires. For the wage equation with IQ as a proxy for ability, express each in words specific to that example.
4. Adding IQ moves the return to education from 6.5% to 5.4%. Explain why the direction of this change is what omitted-variable-bias reasoning predicts.
5. A colleague objects that adding IQ worsens multicollinearity. Give the strongest version of the objection and then the answer to it.
6. Why does including the 1982 crime rate change the coefficient on law enforcement expenditure so much? What experiment does the lagged variable make possible?
7. State the two conditions an instrument must satisfy. Then explain in one sentence why IQ satisfies neither role well when used as an instrument for education.
8. Why can exper serve as its own instrument while educ cannot? What extra variable is needed and what must it satisfy?
9. Write the decomposition educ = êduc + (educ − êduc) and explain which part 2SLS uses and why the other is discarded.
10. The IV return to education is 5.9% against an OLS estimate of 10.8%. What does the direction of the difference suggest, and why is the IV estimate less precise?
11. Explain what the J-statistic tests and why it provides only partial protection against invalid instruments.
12. A Hausman test gives p = 0.09. What do you conclude about whether education is endogenous, and what would you do next?
13. The Cragg-Donald statistic is 55.4. What is being tested, and what would a value of 3 have implied for your estimates?

14. List the main criticisms of IV estimation. Which of them would concern you most in the education example, and why?
15. In the pooled wage equation, what do δ_1 and δ_5 measure? Why must the base year be excluded?
16. Explain why the nominal-to-real wage transformation is unnecessary here, and identify the one coefficient it would change.
17. Why is comparing high and low earners after the Kentucky policy change a naive analysis? What does difference-in-differences add?
18. Write out the four group means used in the DiD estimator and show how they combine. What does each of the two differences remove?
19. The coefficient on afchnge is 0.008 with p = 0.86. Why is this insignificance a point in favour of the analysis rather than against it?

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 9, 13 and 15 cover the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 8.